Experiment 2: Water Jug Problem

Aim

To implement and solve the Water Jug Problem using Artificial Intelligence techniques.

Algorithm

1. Start
2. Initialize the problem state
3. Apply valid operations/search
4. Repeat until goal state is reached
5. Display solution
6. Stop

Flowchart

```
Start
  |
Initialize
  |
Process/Search
  |
Goal Reached?
 /   \
Yes  No
 |    |
Print Repeat
 |
Stop
```

Objective

- Understand state-space search.
- Implement AI search technique.
- Obtain the required goal state.

Program

```
from collections import deque

def water_jug(jug1, jug2, target):

    visited = set()

    queue = deque([(0, 0)])

    while queue:

        x, y = queue.popleft()
```

```
        if (x, y) in visited:
            continue
        print((x, y))
        visited.add((x, y))
        if x == target or y == target:
            print("Target Reached")
            return
        next_states = [
            (jug1, y),                        # Fill Jug 1
            (x, jug2),                        # Fill Jug 2
            (0, y),                         # Empty Jug 1
            (x, 0),                         # Empty Jug 2
            (max(0, x-(jug2-y)), min(jug2, x+y)),  # Pour Jug 1 → Jug 2
            (min(jug1, x+y), max(0, y-(jug1-x)))   # Pour Jug 2 → Jug 1
        ]
        for state in next_states:
            if state not in visited:
                queue.append(state)

water_jug(4, 3, 2)
```

Output

```python
from collections import deque

def water_jug(jug1, jug2, target):
    visited = set()
    queue = deque([(0, 0)])

    while queue:
        x, y = queue.popleft()

        if (x, y) in visited:
            continue

        print((x, y))
        visited.add((x, y))

        if x == target or y == target:
            print("Target Reached")
            return
```

```
(0, 0)
(4, 0)
(0, 3)
(4, 3)
(1, 3)
(3, 0)
(1, 0)
(3, 3)
(0, 1)
(4, 2)
Target Reached

=== Code Execution Successful ===
```

Result

The program was executed successfully.

Conclusion

The program demonstrated the AI concept successfully and produced the expected result.